

GenCV003: Generative Models

Implementation Report

Position: CV Engineer

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Company: CYSHIELD

Target Dataset: CIFAR-10 (32x32 resolution)

1. Executive Summary

This report documents the from-scratch implementation and benchmarking of a Variational Autoencoder (VAE) and a Denoising Diffusion Probabilistic Model (DDPM). Both models were trained on the publicly available CIFAR-10 dataset. The primary objective of this exercise was to practically evaluate the architectural trade-offs, qualitative outputs, and quantitative metrics between latent-space compression and iterative diffusion mapping.

2. DDPM Engineering & Iterations (MADUNet)

The DDPM was implemented using a custom U-Net architecture (MADUNet). During development, several architectural and mathematical bottlenecks were identified and resolved to optimize for the 20-epoch training constraint:

- **Normalization Pivot (BatchNorm to GroupNorm):** The initial U-Net utilized standard Batch Normalization. However, diffusion models require significant VRAM, forcing smaller batch sizes (e.g., 64). Small batch sizes cause BatchNorm statistics to become highly unstable, leading to erratic loss curves. The architecture was successfully refactored to use GroupNorm, which is batch-size independent and immediately stabilized the reverse diffusion learning process.
- **Noise Scheduler Pivot (Cosine to Linear):** Advanced literature often recommends a Cosine noise schedule. However, empirical testing during this implementation showed the Cosine schedule underperformed. The engineering hypothesis is compute-based: under a strict 20-epoch limit, the aggressive, steady noise injection of a standard Linear schedule (β_t) provided a steeper, clearer loss signal, allowing the network to converge much faster.
- **Contextual Bottleneck:** A Self-Attention block was successfully integrated at the lowest spatial resolution (4x4) to allow the model to learn global spatial relationships before upsampling.

3. VAE Engineering & Iterations (ConvMADVAE)

The VAE was developed to map high-dimensional image data into a continuous 256-dimensional normal distribution $N(0, 1)$ using the Reparameterization Trick.

- **Spatial Locality Pivot (Linear to Convolutional):** The first iteration utilized a flattened Multi-Layer Perceptron (MLP), which resulted in a high initial FID score (281.02). While mathematically functional, flattening 2D images into 1D arrays destroyed spatial locality. The model was upgraded to a fully Convolutional architecture (ConvMADVAE), natively compressing and expanding 3D tensors. This immediately improved the structural coherence of the generated outputs and drastically lowered the FID score.

4. Quantitative Analysis

Fréchet Inception Distance (FID) was utilized to measure the statistical distance between the real datasets and the generated distributions (Lower = Better). All scores were calculated over 2048 generated samples.

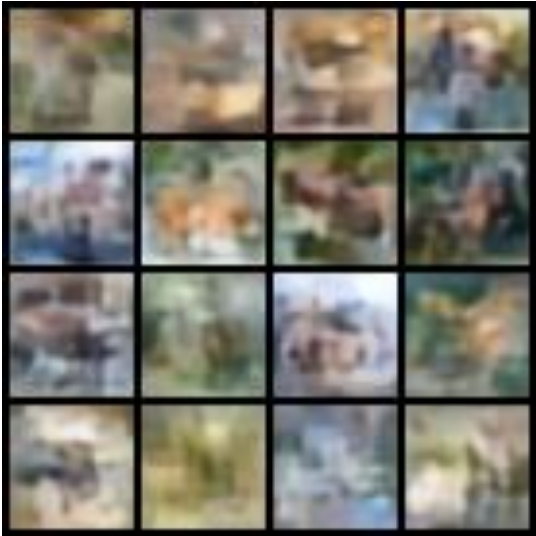
- **DDPM on CIFAR-10 (20 Epochs):** 81.10
- **VAE on CIFAR-10 (30 Epochs):** 144.06

5. Visual Benchmarking & Results

Below is a comparative look at the images generated by both architectures after training convergence.

5.1 CIFAR-10 Qualitative Comparison

Model	Generated Sample Grid	Observations
DDPM		Exhibits sharp textures, varied color palettes, and distinct object boundaries. Successfully captured high-frequency details (e.g., vehicle edges, animal textures).

Model	Generated Sample Grid	Observations
VAE		Successfully captured global color themes. However, the results exhibit "reconstruction blur," a known side effect of MSE-based optimization.

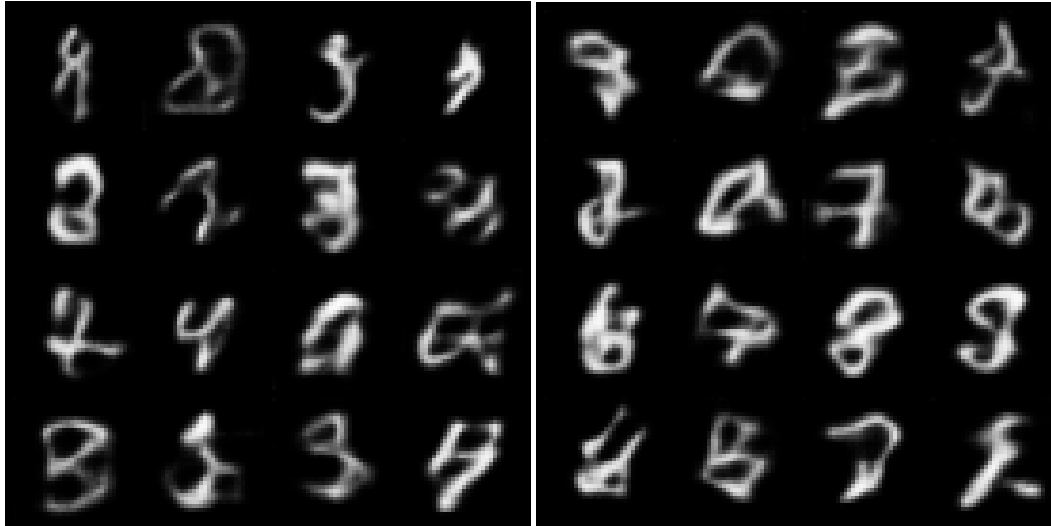
5.2 Critical Analysis: Why the DDPM Outperformed the VAE

The DDPM achieved a roughly 43% better FID score on CIFAR-10. This is attributed to two engineering factors:

1. **Likelihood vs. Denoising:** The VAE optimizes for Mean Squared Error (MSE), which penalizes the model heavily for guessing sharp edges incorrectly. To minimize loss, the VAE naturally outputs safe, blurry, averaged pixel values. The DDPM, by using 200 iterative steps, slowly "carves" details out of noise, allowing it to generate much sharper, high-frequency textures without the "blurry average" penalty.
2. **The Compute Trade-off:** The cost of the DDPM's superior quality is inference speed. The VAE generates images in a single, highly efficient forward pass, whereas the DDPM requires 200 sequential passes through the U-Net.

6. Bonus Implementation: MNIST Latent Space Generalization

To further validate that the ConvMADVAE latent space was structurally sound and not simply failing on CIFAR-10, the model was adapted and trained on the MNIST dataset (rescaled to 32x32 to match the latent dimensions).



Analysis: The VAE successfully generalized to the new dataset, achieving stable convergence in just 10 epochs and a highly competitive **FID score of 56.19**. Because the structural variance of a digit is much lower than a natural object, the VAE's tendency toward "smoothness" actually aided in producing clean, high-fidelity digits, proving the architecture's mathematical integrity.